S3.1 The periodic table: Classification of elements

Discontinuities in ionisation energy (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to explain how discontinuities in first ionisation energy provide evidence for the existence of energy sublevels.

In section S3.1.3b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/) we looked at the general trends in ionisation energy, the energy required to remove an electron. However, we only looked generally at these trends down a group and across a period. **Figure 1** shows the first ionisation energies for elements in periods 1 to 3. Looking at this, does the first ionisation energy always increase as you travel across a period? Are there any elements that stand out as anomalies to this trend? Why do these elements fall outside of the general trend? Why is it important to analyse data that fall outside of general trends?

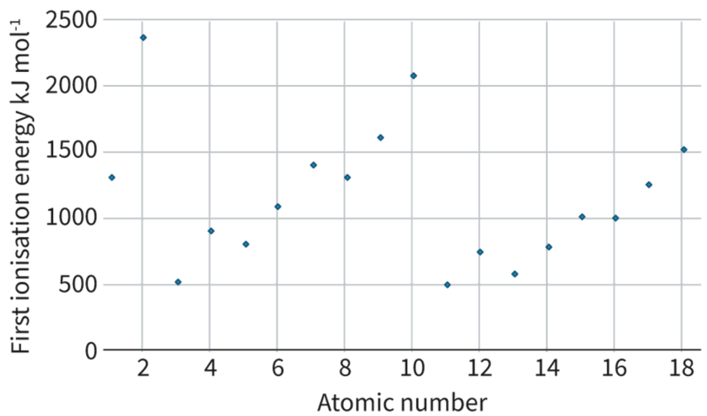

Figure 1. First ionisation energies for elements in periods 1 to 3.

More information for figure 1

The scatter plot illustrates the first ionisation energies for elements in periods 1 to 3 of the periodic table. The X-axis represents the atomic number of the elements, ranging from 0 to 18. The Y-axis represents the first ionisation energy in kilojoules per mole (kJ mol^-1), ranging from 0 to 2500.

Data points indicate the ionisation energies of specific elements. The plot shows a trend of increasing ionisation energy across the periods, but there are noticeable fluctuations and anomalies. Not all elements show a consistent increase in ionisation energy, with some dips and spikes occurring, representing the discontinuities or anomalies in the general trend.

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You may recall from section S1.3.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/) that these anomalies are called discontinuities. **Figure 2** highlights these discontinuities and takes a closer look at what elements fall near them. Looking at **Figure 2**, consider the following questions: What groups do these discontinuities fall between? What are the electron configurations of the valence shell for these elements? Why might these discontinuities exist?

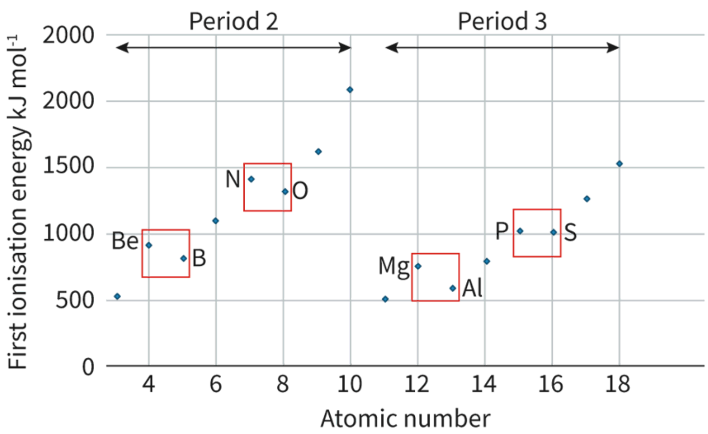

Figure 2. Discontinuities in ionisation energies in periods 2 and 3.

More information for figure 2

The graph shows the first ionisation energy in kJ mol^-1 on the Y-axis, ranging from 0 to 2000, against atomic numbers on the X-axis, ranging from 4 to 20, which correspond to elements across periods 2 and 3 of the periodic table. Data points indicate the ionisation energies for elements like Be, B, N, O, Mg, Al, P, and S. There are highlighted sections with red boxes showing discontinuities between certain elements such as Be and B, N and O, Mg and Al, and P and S. The graph includes horizontal lines marking energy levels and vertical lines indicating atomic numbers, assisting in identifying these discontinuities.

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These discontinuities appear in two distinct regions moving across a period on the periodic table:

- Between groups 2 and 13
- Between groups 15 and 16

These discontinuities observed, based on decreasing ionisation energy values, indicate that it is easier to remove an electron from elements in group 13 than in group 2, and to remove from elements in group 16 than group 15. The lower than expected ionisation energy values suggest that group 13 and group 16 elements have a weaker attraction between the nucleus and outer valence shell electrons than we would expect, given their higher effective nuclear charge than group 12 and group 15 elements. So why might this attraction be weaker? Let's take a closer look at the electron configuration for elements in these groups in **Table 1**.

Table 1. General electron configurations for valence electrons for elem

Period	Outer energy level	Group 2		Group 13		Gr
2	*n* = 2	Be	[He] $2s^2$	B	[He]$2s^2 2p^1$	N
3	*n* = 3	Mg	[Ne] $3s^2$	Al	[Ne]$3s^2 3p^1$	P

First discontinuity

What do you notice about the sublevel that the outer electron occupies in group 13 in comparison to in group 2? Let's look at an orbital diagram to compare the sublevels of the outer electrons for beryllium (Be) and boron (B) in **Figure 3**.

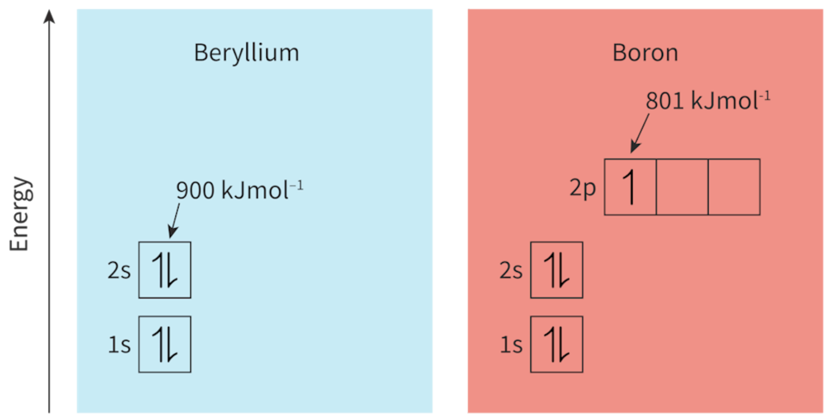


Figure 3. Orbital diagrams for beryllium and boron.

More information for figure 3

The image displays two orbital diagrams side-by-side, comparing the electron configurations of beryllium and boron.

On the left, the diagram for beryllium shows two electron shells. The 1s sublevel contains two electrons with opposite spins, represented as arrows pointing in opposite directions within a box labeled "1s." The 2s sublevel also contains two electrons with opposite spins in a box labeled "2s." The energy level is marked as 900 $kJmol^{-1}$.

On the right, the diagram for boron displays three sublevels. Like beryllium, the 1s and 2s sublevels each contain two electrons with opposite spins in their corresponding boxes. The new 2p sublevel contains one electron in the first of three available slots, represented by a single upward arrow in a box labeled "2p." The energy level is marked as 801 $kJmol^{-1}$.

These diagrams illustrate that beryllium has its outer valence electron in the 2s sublevel, while boron has its outer valence electron entering the higher energy 2p sublevel.

[Generated by AI]

Group 2 elements, like beryllium, have an outer valence electron occupying an s-sublevel, whereas group 13 elements boron have an outer valence electron occupying a p-sublevel. Since p-sublevels are slightly higher in energy than s-sublevels, the nucleus has a weaker attraction to these electrons. This is why the first ionisation energy of elements in group 13 is lower than for elements in group 2 as we move across a period.

This discontinuity observed in the general trend of first ionisation energy provides evidence for the existence of a sublevel that is slightly higher in energy for the third valence electron to occupy.

Second discontinuity

What do you notice about the sublevel that the outer electron occupies in group 15 in comparison to in group 16? Let's look at an orbital diagram to compare the sublevels of the outer electrons for nitrogen (N) and oxygen (O) in **Figure 4**.

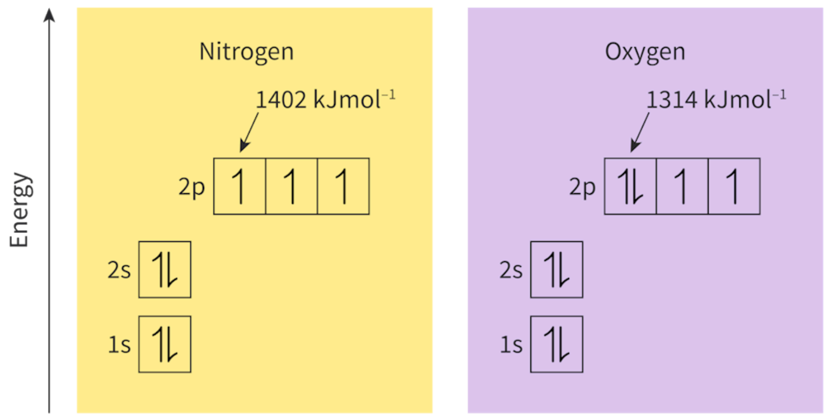


Figure 4. Orbital diagrams for nitrogen and oxygen.

More information for figure 4

The image displays two orbital diagrams comparing the electron configurations of nitrogen and oxygen. On the left, the nitrogen diagram is presented with a yellow background. It shows a vertical axis labeled "Energy." Below are three horizontal boxes labeled 1s, 2s, and 2p. The 1s and 2s orbitals each have two arrows in opposite directions, denoting paired electrons. The 2p sublevel is shown with three boxes, each containing a single arrow, indicating singly occupied orbitals. The text "1402 kJmol^-1" is highlighted, reflecting nitrogen's ionization energy.

On the right, the oxygen diagram is on a purple background. It also includes three horizontal boxes labeled 1s, 2s, and 2p. Similar to nitrogen, the 1s and 2s orbitals are fully occupied with paired arrows. However, in the 2p sublevel, the first box has two arrows in opposite directions, noting double occupancy, while the other two boxes contain a single arrow each. The ionization energy for oxygen is specified as "1314 kJmol^-1."

This comparison illustrates the difference in electron configuration between the two elements, particularly in the 2p sublevel where nitrogen has singly occupied orbitals, whereas oxygen has one doubly occupied orbital.

[Generated by AI]

Both of these elements have outer valence electrons occupying a p-sublevel. Group 15 elements, such as nitrogen, have valence electrons singly occupying each orbital within the p-sublevel, whereas group 16 elements, such as oxygen, have one orbital in the p-sublevel with double occupancy. Due to the orbital's double occupancy, there will be added repulsion between the two electrons in the same orbital. This increased repulsion decreases the attraction between the valence electrons and the nucleus making it easier to remove an electron. Hence, this explains the slightly lower first ionisation energy of elements in group 16 in comparison to group 15.

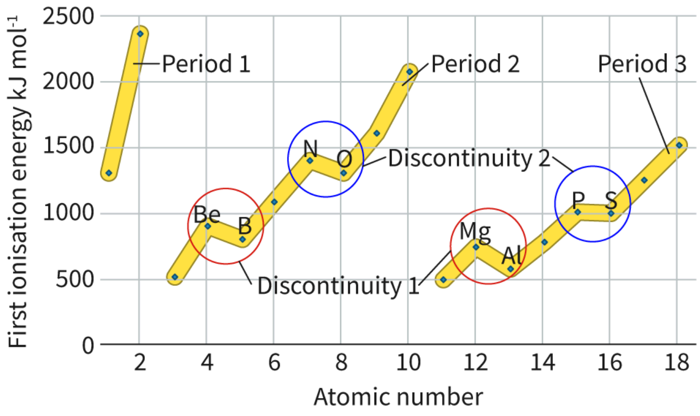
This discontinuity observed in the general trend of first ionisation energy provides evidence for the existence of orbitals able to hold a maximum of two electrons that make up each sublevel.

Nature of Science

Aspect: Evidence

Since atomic orbitals are just a region of space where there is a high probability of finding an electron, evidence of their existence is quite limited (i.e. we cannot just see an orbital with a microscope). First ionisation energies are quantitative data that we can measure. Trends in ionisation energies, along with their discontinuities provide a lot of evidence to show how electrons are arranged within an atom. The general trend in ionisation energy provides evidence for the existence of energy levels, but it is the discontinuities that provide evidence for the existence of sublevels and atomic orbitals.

Interactive 1 is an animation showing a closer view of these discontinuities, by looking at the way that electrons are arranged within their sublevels.



Interactive 1. A Closer Look at the Discontinuities of First Ionisation Energy.

More information for interactive 1

An interactive slideshow with five slides presents the discontinuity of the first ionisation energy across elements. Each slide presents a scatterplot exhibiting first ionisation energy in \(\mathrm{KJmol}^{-1}\) on the y-axis and atomic number on the x-axis. The y-axis ranges from 0 to 2500 \(\mathrm{KJmol}^{-1}\), and the x-axis ranges from 0 to 18.
Slide 1:
This slide depicts ionisation energy trends for Periods 1, 2, and 3. In each period, a line connects the data points for its elements. The following table presents the approximated data:

All periods show an overall increasing trend, with the discontinuities in Periods 2 and 3.
Discontinuity 1:

Atomic number	First ionisation energy ($KJmol^{-1}$)
Period 1	
1	1350
2	2400
Period 2	
3	500
4	900
5	850
6	1150
7	1400
8	1300
9	1650
10	2100
Period 3	
11	500
12	750
13	600
14	800
15	1050
16	1000
17	1250
18	1500

Beryllium, Be (Atomic number \(= 4\) and Ionisation energy \(= 900\)) and Boron, B (Atomic number \(= 5\) and Ionisation energy \(= 850\)). Boron has a lower ionisation energy than Beryllium.

Magnesium, Mg (Atomic number $= 12$ and Ionisation energy $= 750$) and Aluminum, Al (Atomic number $= 13$ and Ionisation energy $= 600$). Aluminum has a lower ionisation energy than Magnesium.
Discontinuity 2:
Nitrogen, N (Atomic number $= 7$ and Ionisation energy $= 1400$) and Oxygen, O (Atomic number $= 8$ and Ionisation energy $= 1300$). Oxygen has a lower ionisation energy than Nitrogen.
Phosphorus, P (Atomic number $= 15$ and Ionisation energy $= 1050$) and Sulfur, S (Atomic number $= 16$ and Ionisation energy $= 1000$). Sulfur has a lower ionisation energy than Phosphorus.
Slide 2:
This slide highlights discontinuity 1 in detail.
Beryllium and Magnesium are from group 2, and Boron and Aluminum are from group 13. Their electronic configuration and their orbital filling are as follows:
Group 2 elements:
Be: $1\mathrm s^2\;2\mathrm s^2$
1s orbital has 2 electrons, 2s orbital has 2 electrons, whereas 2p, 3s, and 3p orbitals are empty.
Mg: $1\mathrm s^2\;2\mathrm s^2\;2\mathrm p^6\;3\mathrm s^2$
1s, 2s, and 3s orbitals each have 2 electrons and 2p orbital has six electrons, whereas 3p orbital is empty.
Group 13 elements:
B: $1\mathrm s^2\;2\mathrm s^2\;2\mathrm p^1$
1s and 2s orbitals each have 2 electrons and 2p orbital has one electron, whereas 3s and 3p orbitals are empty.
Al: $1\mathrm s^2\;2\mathrm s^2\;2\mathrm p^6\;3\mathrm s^2\;3\mathrm p^1$
1s, 2s, and 3s orbitals each have 2 electrons, 2p orbital has six electrons, and 3p orbital has one electron.
These valence electrons are slightly higher in energy than the valence electrons in group 2 elements.
Slide 3:
Therefore electrons from Boron and Aluminum are easier to remove than the outer electron from group 2 elements.
Slide 4:
This slide highlights discontinuity 2 in detail.
Nitrogen and Phosphorus are from group 15, and Oxygen and Sulfur are from group 16. Their electronic configuration and their orbital filling are as follows:
Group 15 elements:
N: $1\mathrm s^2\;2\mathrm s^2\;2\mathrm p^3$
1s and 2s orbitals each have 2 electrons and 2p orbital has 3 electrons, whereas 3s and 3p orbitals are empty.
P: $1\mathrm s^2\;2\mathrm s^2\;2\mathrm p^6\;3\mathrm s^2\;3\mathrm p^3$
1s, 2s, and 3s orbitals each have 2 electrons, 2p orbital has six electrons, and 3p orbital has 3 electrons.
Only single occupancy.
Group 16 elements:
O: $1\mathrm s^2\;2\mathrm s^2\;2\mathrm p^4$
1s and 2s orbitals each have 2 electrons and 2p orbital has four electrons, whereas 3s and 3p orbitals are empty.
S: $1\mathrm s^2\;2\mathrm s^2\;2\mathrm p^6\;3\mathrm s^2\;3\mathrm p^4$
1s, 2s, and 3s orbitals each have 2 electrons, 2p orbital has six electrons, and 3p orbital has four electrons.
Double occupancy causes repulsion of electrons in the orbital.
Slide 5:
Repulsion between the electrons from Oxygen and Sulfur reduces attraction between nucleus and electrons. Easier to remove and lower IE.
This slideshow helps users identify the periodic trends, relate electronic

configuration and orbital occupancy, and recognise the discontinuities in ionisation energy.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Thinking — Asking questions and framing hypotheses based upon sensible scientific rationale
- **Time required to complete activity:** 45 minutes
- **Activity type:** Group activity

Model of the atom mystery box

Materials needed per group:

- Box (shoebox size or smaller)
- 5 marbles
- Assortment of craft materials (cardboard, paper, toilet paper rolls, felt, glue, stapler, tape, etc.)

Creating a model of the atom (groups of 3—4 people)

- Your task is to create a model of an atom with 5 electrons, these electrons will be represented by marbles.
- You are going to come up with your own model of how these electrons are arranged in your atom. Your atom will be represented by the inside of the box.
- Your electrons could be fixed, move in an orbit, in a line, have free motion over certain regions of the atom (box). The choice is yours.
- You will use the craft materials you have at your disposal to design the arrangement of electrons within your atom.
- When complete your box will be sealed shut. Your peers will have to use evidence they can collect by moving, shaking and listening to come up with their own theory about how atoms are arranged in other students' boxes.

Creating models using evidence

- Swap your box with another group's box.
- Make observations (moving, listening, light shaking, rocking side to side) to try and figure out how the 5 electrons are arranged within the atom.
- Create a sketch to show how you think the electrons are arranged. Note all the evidence you have to support this arrangement.

Sharing models with class

- Each group will present their sketches to the class for how they think electrons are arranged in other group's boxes, making explicit mention of what evidence they have to support each part of their sketch.

- Students can then decide whether to reveal what the inside of the original box looked like or not, keeping in mind that we never really have the opportunity to see inside the atom to confirm the models we construct from evidence.

Class discussion

- Discuss how this activity relates to:
  - Models of the atom.
  - The importance of using different types of evidence when constructing models, particularly when visual observations are extremely limited.